EXTENSION BY ZERO COMMUTES WITH ETALE RESTRICTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-restrict-lower-shriek; the proof environment immediately following it is its own proof

Editorial note. The label lemma-restrict-lower-shriek names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the extension by zero functors j_! for an etale morphism j : U to X of algebraic spaces, constructed termwise on complexes of O_X-modules (sites-modules-lemma-extension-by-zero of Modules on Sites)

Source reference. Original lemma 03DI: sources/stacks-chow/tags.txt, source lines 4421–4421. Prerequisite pointer invoked at proof line(s) 2131; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $j : U \to X$ be a étale morphism of algebraic spaces over S. Given an étale morphism $V \to Y$, set $W = V \times_X U$ and denote $j_W : W \to V$ the projection morphism. Then $(j_!E)|_V = j_{W!}(E|_W)$ for E in $D(\mathcal{O}_U)$.*

Proof. This is true because $(j_!\mathcal{F})|_V = j_{W!}(\mathcal{F}|_W)$ for an $\mathcal{O}_X$-module $\mathcal{F}$ as follows immediately from the construction of the functors $j_!$ and $j_{W!}$, see Modules on Sites, Lemma 03DI. □

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